

TSOGT-OVOO, Mongolia

WMO: 443470

Lat: 44.4232N

Lon: 105.3180E

Elev: 1298

StdP: 86.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-24.8	-31.3	0.2	-24.0	-29.3	0.3	-22.5	14.5	-9.7	13.7	-8.7	1.4	320	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.3	16.6	31.5	15.9	29.9	15.5	22.1	25.4	19.6	25.1	17.8	25.5	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	18.6	23.7	18.1	15.3	21.7	15.4	12.9	20.3	72.4	25.2	62.4	24.8	55.7	25.6	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.3	10.4		-29.3	36.0	2.8	1.9	-31.3	37.4	-32.9	38.5	-34.5	39.6	-36.5	40.9
				-29.6	20.6	2.6	3.0	-31.5	22.8	-33.1	24.6	-34.5	26.3	-36.4	28.5
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-15.3	-10.8	-2.1	7.3	13.9	20.2	23.3	21.2	14.7	5.4	-4.9	-13.0	(6)
(7)		DBStd	14.25	5.02	5.77	6.07	5.31	4.99	3.87	3.37	3.50	4.63	5.11	5.72	4.84	(7)
(8)		HDD10.0	3207	785	582	375	112	21	0	0	0	13	157	447	714	(8)
(9)		HDD18.3	5212	1043	816	633	331	153	24	3	13	124	402	697	972	(9)
(10)		CDD10.0	1407	0	0	1	31	141	306	412	348	154	13	0	0	(10)
(11)		CDD18.3	372	0	0	0	1	14	81	157	103	16	0	0	0	(11)
(12)		CDH23.3	3858	0	0	0	8	216	882	1652	942	157	1	0	0	(12)
(13)		CDH26.7	1398	0	0	0	1	52	308	694	318	26	0	0	0	(13)
(14)	Wind	WSAvg	4.3	3.4	3.7	4.8	5.5	5.6	4.6	4.1	4.0	4.1	4.2	3.5	(14)	
(15)	Precipitation	PrecAvg	95	1	1	2	3	8	16	30	24	9	4	2	1	(15)
(16)		PrecMax	211	3	11	8	22	42	46	90	89	51	17	8	10	(16)
(17)		PrecMin	49	0	0	0	0	0	0	1	0	0	0	0	0	(17)
(18)		PrecStd	39	1	2	2	5	8	11	26	19	10	4	2	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	6.3	16.3	24.8	30.8	34.2	36.9	34.2	29.3	21.8	11.9	1.1	(19)
(20)		MCWB	-4.1	-0.4	5.4	10.7	13.3	16.1	19.5	16.6	15.0	9.8	4.7	-3.2	(20)	
(21)		2%	DB	-3.3	3.1	13.3	22.1	28.1	32.0	34.2	31.8	27.1	18.6	8.6	-1.7	(21)
(22)		MCWB	-6.1	-1.8	4.0	9.2	12.9	15.4	17.2	15.9	13.2	7.8	2.1	-5.0	(22)	
(23)		5%	DB	-5.7	0.8	10.4	19.5	25.9	30.0	32.2	30.0	25.1	16.2	6.1	-4.0	(23)
(24)		MCWB	-8.1	-3.4	2.3	7.8	11.5	14.9	16.4	15.9	12.2	6.7	0.7	-6.7	(24)	
(25)		10%	DB	-7.7	-1.6	7.6	17.0	23.6	28.2	30.5	28.2	22.9	13.8	3.7	-5.9	(25)
(26)		MCWB	-9.6	-4.9	0.9	6.6	10.7	13.8	16.0	15.0	11.4	5.5	-0.6	-8.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.6	0.9	7.9	13.9	20.3	25.3	25.9	24.3	19.0	11.5	5.8	-2.8	(27)
(28)		MCDWB	-0.7	4.8	12.2	18.5	23.2	28.4	28.0	27.1	24.0	18.4	9.3	0.5	(28)	
(29)		2%	WB	-6.0	-1.4	4.6	10.6	16.4	21.2	22.2	20.4	15.6	8.8	3.1	-4.8	(29)
(30)		MCDWB	-3.3	2.7	11.8	18.5	22.7	23.4	26.7	24.5	22.4	16.5	7.9	-2.0	(30)	
(31)		5%	WB	-7.9	-3.1	2.7	8.5	13.3	17.4	19.7	18.3	13.6	7.3	1.2	-6.5	(31)
(32)		MCDWB	-5.7	0.7	10.2	18.2	22.4	24.1	26.0	25.3	21.9	15.3	6.1	-4.1	(32)	
(33)		10%	WB	-9.5	-4.8	0.9	6.9	11.4	15.6	18.0	16.8	12.1	5.9	-0.3	-8.0	(33)
(34)		MCDWB	-7.6	-1.6	7.6	16.5	22.2	24.8	26.4	24.6	21.0	13.3	4.1	-5.9	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.6	12.4	12.8	12.7	12.7	11.8	11.1	10.8	11.7	11.8	10.9	10.1	(35)	
(36)		5% DB	MCDBR	12.3	15.0	15.4	15.4	15.2	13.6	12.9	12.7	13.6	14.5	13.2	11.8	(36)
(37)		MCWBR	10.0	11.1	9.5	8.5	7.6	5.8	4.7	5.0	6.3	8.1	8.7	9.3	(37)	
(38)		5% WB	MCDWB	12.1	14.6	14.9	14.8	14.1	11.8	10.9	10.7	12.3	13.4	12.6	11.6	(38)
(39)		MCWBR	10.0	11.0	10.0	9.5	9.4	7.8	6.3	6.0	7.0	8.2	8.9	9.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.286	0.328	0.373	0.393	0.412	0.393	0.375	0.340	0.305	0.273	0.257	(40)	
(41)		taud	2.327	2.265	2.178	2.102	2.096	2.126	2.247	2.275	2.321	2.352	2.354	2.325	(41)	
(42)		Ebn at Noon	871	903	901	885	875	858	871	875	880	869	844	833	(42)	
(43)		Edh at Noon	78	101	127	150	156	152	133	125	109	92	76	71	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	3.18	4.84	6.15	7.09	7.01	6.65	6.10	5.14	3.81	2.47	1.81	(44)	
(45)		RadStd	0.28	0.39	0.19	0.27	0.34	0.35	0.30	0.24	0.19	0.13	0.21	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-2.20	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-2.21	N/A	-1.18	-1.21	N/A	N/A	N/A	N/A

Nomenclature: See separate page